

DAA Assignment April 2026

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1. To prove that the following problems are in NP-complete, show the reductions for the following problems:
 - (a) SAT $\rightarrow$ 3SAT
 - (b) Independent Set $\rightarrow$ Vertex Cover(Note that in the Independent Set problem we are given a graph and an integer g , and the aim is to find g vertices that are independent, that is, no two of which have an edge between them. The Vertex Cover problem considers a graph and a budget b , and the idea is to find b vertices that cover (touch) every edge.)
 2. A large region of cities is connected by a network of roads, forming a map $G = (V, E)$. Each road has a travel time based on its length and condition. Two major cities, s (the capital) and t (a key industrial hub), suffer from heavy traffic because the fastest existing route between them is still too slow. The government has funding to build exactly one new highway, but due to budget and terrain constraints, it can only be built between certain pairs of cities. These possible new highways are listed in a set E' , each with an estimated construction length (and thus travel time). As the chief planner, your job is to decide which single new highway should be built so that travel time between the capital s and the industrial city t is reduced as much as possible. Give an efficient algorithm for solving this problem and prove its correctness.
 3. We had seen the divide and conquer based approach to find the k^{th} smallest element in an array where we picked a pivot randomly. However, one can see that the worst case complexity of this approach would be $O(n^2)$. However there is an alternative solution called "Median of Medians" that continues to rely on the technique of divide and conquer but picks a pivot using a deterministic approach. State the algorithm and justify why this approach has a better run time complexity even in the worst case.